

PlayIT: Offline-First Gamified Early Literacy Mobile Application

MVP Validation Framework & SMART Evaluation Model (UPA Model — Weeks 1–2 Deliverable)

Course: IT411 — Capstone & Research 2 | Semester 1, AY 2026–2027
Institution: College of Computer Studies, Cebu Institute of Technology – University
Primary Framework: UPA (Usability, Pedagogy, Accessibility) | Standard SUS + Child Smileometer
Date of Submission: September 13, 2026 | Document Version: 2.0 (Aligned with Google Form Package)

1. Research Rationale & Validation Objective

Early literacy in Philippine public schools faces high pupil-to-teacher ratios and limited home internet access. The Department of Education (DepEd) mandates the **Marungko Approach**—a phono-syllabic reading method that introduces letters by phonemic frequency. **PlayIT** provides home-based phonics practice through a 100% offline Android app with Vosk speech recognition, gamification, and diagnostic telemetry.

The Weeks 1–2 MVP Validation is **formative and diagnostic**: (1) evaluate pediatric usability and child task flow; (2) establish baseline reading gain via empirical pre/post assessment; (3) obtain DepEd teacher validation on the 26-letter English Marungko progression and 33 CVC Blend It words; and (4) verify offline software resilience before full feature lock.

2. Primary Theoretical Model: The UPA Evaluation Framework

In educational technology for early learners, generic UI heuristics are insufficient. PlayIT adopts the **UPA Framework (Usability, Pedagogy, Accessibility)** as its core evaluation architecture, operationalized across all stakeholder cohorts:

- **Usability (U):** Evaluates navigation ease, interaction efficiency, and user satisfaction. Operationalized via the **Child Smileometer** (3-point visual scale — Read & MacFarlane, 2006) for young learners, the standardized **10-Item System Usability Scale (SUS — Brooke, 1996)** for parents and teachers, and observational task timings (time-on-task, ASR latency).
- **Pedagogy (P):** Evaluates curriculum alignment, phonetic accuracy, and reading efficacy. Operationalized via the **DepEd Teacher Pedagogy Checklist** (7 Likert items covering sequence fidelity, phonemic modeling, and decodable CVC words) and the **Child Pre/Post Phonics Assessment** measuring empirical letter sound and word-blending gain.
- **Accessibility (A):** Evaluates physical ergonomics and socioeconomic equity. Operationalized via **Pediatric Touch-Target Standards** (>=64dp buttons for motor control), high-contrast visual cues (WCAG 2.1 AA), audio-visual dual-coding, and **100% Offline Zero-Data Operability** with zero ads or tracking.

3. Curriculum Scope: 26-Letter Marungko English Adaptation

PlayIT adapts the 28-letter Filipino Marungko sequence to **26 letters (A–Z)** for Grade 1 English phonics. The Spanish-origin letter **Ñ** is excluded because it does not exist in English. The digraph **NG (/ŋ/)** is excluded from individual graphemes because teaching it in isolation creates phonological confusion for early readers. The 26 letters are organized into **7 progressive chapters** with **33 decodable CVC Blend It words**:

Chapter	Target Letters	Count	Biome Theme	Canonical Blend It Words (CVC Validated)
Chapter 1	m, s, a, i	4	Guava Greenery	SAM, SIS, AIM (3 words — Group 1 exception)
Chapter 2	o, b, e, u	4	Mango Orchard	BUS, SUB, MOM, BEE, BIB
Chapter 3	t, k, l, y	4	Chocolate Hills	BAT, MAT, KIT, TOY, BOY
Chapter 4	n, g, p	3	Palm Valley	PIG, PAN, BUG, PIN, NAP
Chapter 5	r, d, h, w	4	Rice Terraces	DOG, HAT, HEN, BED, WEB
Chapter 6	c, f, j	3	Coral Reef	CAT, FAN, CAP, CUP, JAM
Chapter 7	q, v, x, z	4	Mt. Pulag Summit	VAN, BOX, FOX, ZOO, QUIZ

All 33 words are constraint-validated against cumulative letter availability. Audio pronunciations (.mp3) and illustrations (.png) are 100% verified on-disk and seeded in Room DB v3.

4. SMART Evaluation Objectives & Measurable Metrics

SMART Dimension	Project Objective	Evaluation Metric & Instrument	Target Criteria	Cohort
Specific (S)	Verify child task flow and vocal production accuracy.	Sublevel completion rate; 1st-attempt ASR vocal accuracy; time-on-task.	>=85% completion rate; >=75% ASR accuracy.	Early Learners (N=12)
Measurable (M)	Quantify usability and caregiver acceptance using standardized scales.	10-Item SUS score; Child Smileometer; TAM constructs (PU, PEOU, BI).	Mean SUS >=75.0 (Grade B+); Smileometer >=2.6/3.0; TAM PU >=4.2/5.0.	Parents & Teachers (N=15)
Achievable (A)	Validate curriculum alignment and measure empirical learning gain.	Teacher Pedagogy Checklist (7 items); Child Pre/Post Phonics Assessment.	100% agreement (>=4.0/5.0) on Marungko sequence; Significant score gain.	DepEd Teachers (N=5) + Learners
Relevant (R)	Ensure 100% offline persistence, data privacy, and diagnostic reporting.	Room DB persistence audit; PDF export time; Noise gate threshold test.	0% data loss on force-close; PDF export <=3.0s; Noise alert active at >40dB.	Technical Evaluators (N=3)
Time-Bound (T)	Execute validation trials, analyze data, and refactor SRS/SDD.	Milestones mapped to IT411 course schedule.	Validation Protocol: Sept 13; User Trials: Sept 18; Refactored SRS/SDD: Sept 19.	Research Team (Total N=30)

5. Stakeholder Mapping & Questionnaire Item Cross-Walk (UPA Model)

Code	Evaluation Criterion / Question Item	UPA Construct	Target Respondent	Scale
PED-01	Strictly follows DepEd Marungko frequency-based progression.	Pedagogy: Sequence	Teachers / SMEs	1–5 Likert
PED-02	Hear It delivers clear, natural phoneme sound modeling.	Pedagogy: Audio Model	Teachers / SMEs	1–5 Likert
PED-03	Say It uses age-appropriate example words (Mouse, Sun).	Pedagogy: Vocalization	Teachers / SMEs	1–5 Likert
PED-04	Find It uses culturally familiar, unambiguous illustrations.	Pedagogy: Visual Discrim.	Teachers / SMEs	1–5 Likert
PED-05	Blend It uses decodable CVC words respecting cumulative letters.	Pedagogy: Blending	Teachers / SMEs	1–5 Likert
PED-06	Excluding NG and Ñ for 26-letter English phonics is pedagogically sound.	Pedagogy: Scope Validity	Teachers / SMEs	1–5 Likert
PED-07	Non-punitive gamification (hearts/stars) reinforces mastery resilience.	Pedagogy: Motivation	Teachers / SMEs	1–5 Likert
PRE-POST	Child phoneme identification and CVC blending diagnostic test.	Pedagogy: Learning Gain	Early Learners (Tested)	0–10 Score Delta
SUS 1–10	Standard 10-Item System Usability Scale (Brooke, 1996).	Usability: Standard SUS	Parents & Teachers	1–5 Likert (0–100)
SMILEY	Child Smileometer affective reaction (Sad=1, Okay=2, Happy=3).	Usability: Affective	Early Learners	3-pt Visual
OBS 1–5	Observational task completion & time-on-task (Map, Hear, Say, Find, Blend).	Usability: Efficiency	Child (Observed)	3-pt + Seconds
ACC-01	Pediatric touch targets (>=64dp) prevent miss-taps for small hands.	Accessibility: Ergonomics	Parents / Teachers	1–5 Likert
ACC-02	High-contrast visual cues & audio dual-coding support emerging readers.	Accessibility: WCAG AA	Parents / Teachers	1–5 Likert
ACC-03	100% offline access ensures equal opportunity with zero mobile data cost.	Accessibility: Equity	Parents / Guardians	1–5 Likert
ACC-04	Ad-free, zero in-app purchase environment guarantees child privacy & safety.	Accessibility: Safety	Parents / Guardians	1–5 Likert
SYS 1–3	Technical verification: ASR latency <=0.5s, 60 FPS UI, Room DB integrity.	Technical: ISO 25010	Technical Evaluators	Pass / Fail / ms

6. Validation Protocol & Thesis Chapter 4 Integration

Field trials will be conducted in Week 2 using playit-debug.apk on 5 Android devices. Observations, Pre/Post scores, and Google Form submissions will be analyzed using descriptive statistics (means, standard deviations, SUS percentile scores) and inferential tests (Paired t-test on Pre/Post scores). These empirical findings will substantiate the **SRS v3.0 Requirements Traceability Matrix (RTM)** and **SDD v2.0** refactoring due in Week 3.